QAOA for MaxCut

Quantum Approximate Optimization on near-term hardware

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Physics 565 / 656 · Spring 2026

Introduction

Definition of a graph

Vertex set

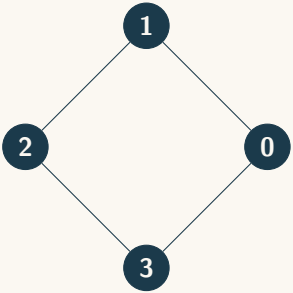
$V = \{0, 1, 2, 3\}, \quad |V| = n = 4$

Edge set

$E \subseteq \binom{V}{2} = \{\{0,1\}, \{1,2\}, \{2,3\}, \{0,3\}\}$

Notation

Throughout: $n = |V|$ qubits, $m = |E|$ edges.
Cuts will be encoded as bitstrings $z \in \{0,1\}^n$.

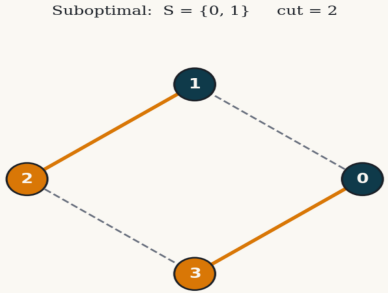


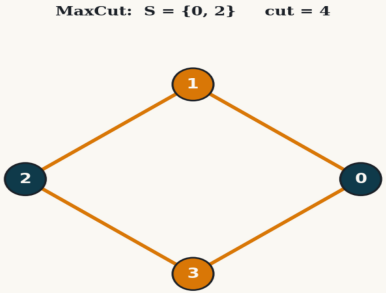
C_4 : cycle on $n = 4$ vertices, $|E| = 4$

The MaxCut problem

$$C(S) \;=\; \big|\{(i,j) \in E : i \in S,\; j \in \overline{S}\}\big|, \qquad S^\star \in \arg\max_{S \subseteq V} C(S).$$

Partition $V = S \sqcup \overline{S}$ to maximise the number of cut edges





For C_4: 2^4 = 16 labelled partitions / 2^3 = 8 distinct cuts. The maximum is 4.

Motivation

QUBO and why MaxCut is a special case

General QUBO. Optimise

$$f(\boldsymbol{x}) \;=\; \sum_{i} Q_{ii}\, x_i \;+\; \sum_{i<j} Q_{ij}\, x_i x_j, \qquad x_i \in \{0,1\}.$$

Encoding MaxCut as QUBO. Let $x_i = 1$ iff vertex $i \in S$. An edge (i,j) is cut iff exactly one endpoint is in S:

$$\mathbf{1}[\mathrm{cut}_{ij}] \;=\; x_i(1-x_j) + (1-x_i)x_j \;=\; x_i + x_j - 2\,x_i x_j.$$

$$C(\boldsymbol{x}) \;=\; \sum_{(i,j)\in E} \big(x_i + x_j - 2\,x_i x_j\big) \;=\; \sum_{i} \deg(i)\, x_i \;-\; 2\sum_{(i,j)\in E} x_i x_j.$$

So MaxCut is QUBO with

$Q_{ii} = \deg(i), \quad Q_{ij} = -2$ if $(i,j) \in E, \quad 0$ otherwise.

Same recipe encodes job scheduling, VLSI layout, Ising ground states, traffic flow, ... — every QUBO is essentially a (weighted) cut problem.

A worked example: portfolio diversification (1/2)

The investor's problem.

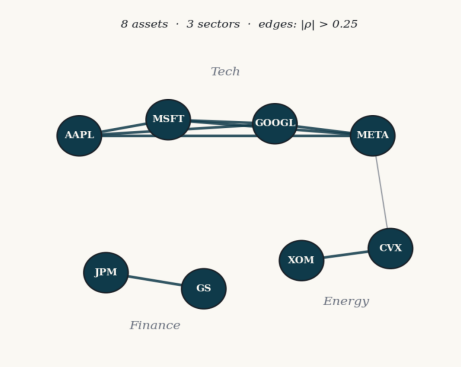
Hold n assets with daily returns $r_i(t)$. *Risk concentrates when assets move together* — so split the portfolio into two halves $S, \overline{S}$, each *internally uncorrelated*, so a shock to one asset does not drag the rest of its half with it.

Measure correlation.

$$\rho_{ij} = \frac{\mathrm{Cov}(r_i, r_j)}{\sigma_i \, \sigma_j} \in [-1, 1]$$

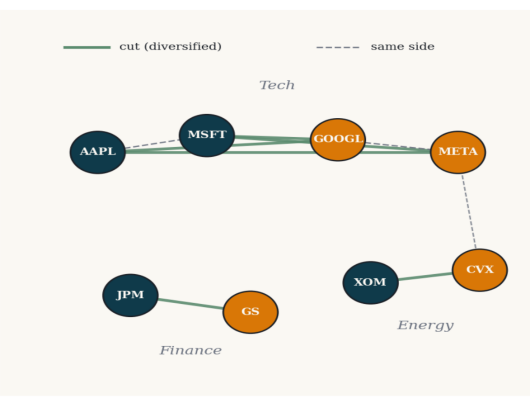
Build the correlation graph.

- Vertex $\leftrightarrow$ asset.
- Edge (i, j) if $|\rho_{ij}| > 0.25$.
- Same-sector pairs $\Rightarrow$ thick edges ($\rho \approx 0.87$).



8 assets, 3 sectors — intra-sector edges dominate.

A worked example: portfolio diversification (2/2)



Cut edges (green) separate correlated pairs; same-side edges (dashed) are correlations within a group.

Why MaxCut works here.
Each *cut edge* separates a correlated pair into different groups. Maximising the cut = separating as many correlated pairs as possible = minimising correlation *inside* each group.

Result on the 8-asset example.
Group S = {AAPL, MSFT, JPM, XOM}, Group $\overline{S}$ = {GOOGL, META, GS, CVX}. Each group contains one asset from every sector — automatic sector diversification, without the investor specifying sectors.

Strategy	Avg. $\lvert\rho\rvert$ within group
Naïve (Tech | Fin+Energy)	0.87 / 0.32
MaxCut	**0.22 / 0.26**

$\sim 4\times$ less correlated inside each group.

Theory

From bitstring to Hamiltonian

1. Decision variable

$z_i \in \{0,1\}$ with rule $z_i = 0 \Leftrightarrow i \in S$, $z_i = 1 \Leftrightarrow i \in \overline{S}$.

Cut size: $C(z) = \sum_{(i,j)\in E} \mathbf{1}[z_i \neq z_j]$.

2. Rewrite with spins

$s_i = 1 - 2z_i \in \{+1, -1\}$:

$$\mathbf{1}[z_i \neq z_j] \;=\; \tfrac{1}{2}(1 - s_i s_j).$$

3. Quantise $s_i \to Z_i$ (Pauli-Z on qubit i):

$$\boxed{H_C \;=\; \sum_{(i,j)\in E} \tfrac{1}{2}(I - Z_i Z_j)}$$

Key property

H_C is *diagonal* in the computational basis:

$$H_C \,|z\rangle = C(z)\,|z\rangle\,.$$

So $\langle H_C \rangle$ is the expected cut value, and the ground states are the MaxCut bitstrings.

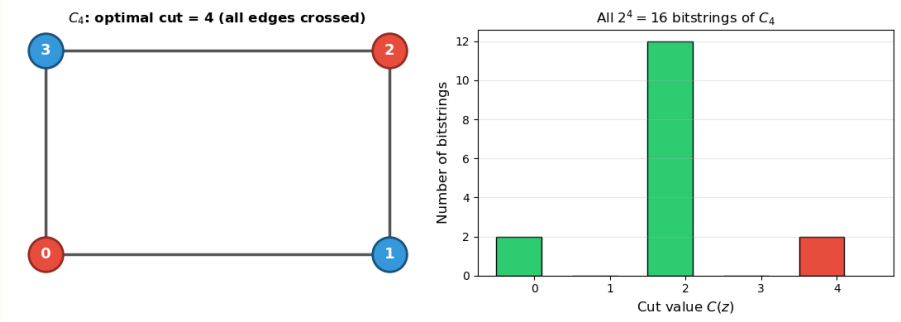
Goal

Prepare $|\psi\rangle$ that maximises $\langle\psi|H_C|\psi\rangle$.

What we are looking for

Measurement rule

Measuring $|\psi\rangle$ in the computational basis returns $z = z_0 z_1 \cdots z_{n-1}$. Read out:
$z_i = 0 \Rightarrow i \in S, \;\; z_i = 1 \Rightarrow i \in \overline{S}$.



Goal: maximise the probability of sampling $|1010\rangle$ or $|0101\rangle$ — the two MaxCut solutions of C_4.

Starting state: $|+\rangle^{\otimes n}$

$$|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle = H^{\otimes n} |0\rangle^{\otimes n}.$$

1. Uniform over candidates. Equal weight on all 2^n bitstrings — no partition is preferred a priori.

2. Aligned with the mixer. $X|+\rangle = +|+\rangle$, so $H_B = \sum_k X_k$ has $|+\rangle^{\otimes n}$ as its top eigenstate.

3. Cheap to prepare. One Hadamard per qubit — constant-depth circuit from $|0\rangle^{\otimes n}$.

4. Adiabatic starting point. Ground state of $H(t=0) = H_B$ in the adiabatic interpolation $H(t) = (1 - t/T)H_B + (t/T)H_C$.

The QAOA ansatz

State

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle \;=\; U_B(\beta_p)U_C(\gamma_p)\cdots U_B(\beta_1)U_C(\gamma_1)\;|+\rangle^{\otimes n}$$

Cost unitary

$$U_C(\gamma) = e^{-i\gamma H_C} \;=\; \prod_{(i,j)\in E} e^{-i\gamma(I - Z_i Z_j)/2}$$

Encodes problem structure — entangling.

Mixer unitary

$$U_B(\beta) = e^{-i\beta H_B}, \quad H_B = \sum_k X_k$$

Drives transitions between bitstrings — single-qubit only.

- Depth $p \;\Rightarrow\; 2p$ variational parameters.
- Larger p: more expressive, but harder to optimise and noisier.

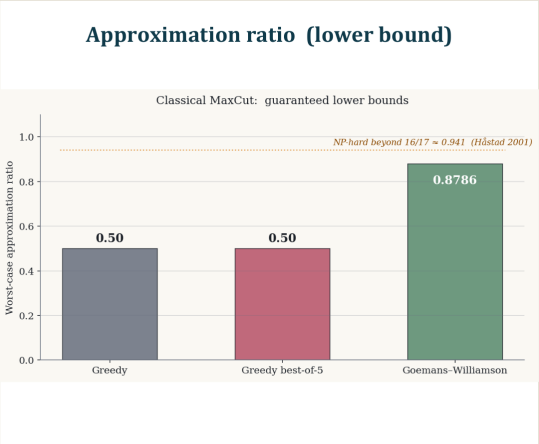
Classical benchmarks: greedy and Goemans–Williamson

Greedy. Process vertices in order; each joins the side cutting more already-assigned edges. Worst-case $r = 1/2$.

Greedy best-of-5. Run greedy 5 times with different orderings; keep the best cut.

Goemans–Williamson (1995). Relax to unit vectors, solve an SDP, round by a random hyperplane. Worst-case $r \geq 0.8786$, runtime $O(n^{3.5})$.

NP-hardness wall. Achieving $r > 16/17 \approx 0.941$ in polynomial time would imply $\mathrm{P} = \mathrm{NP}$ (Håstad 2001).



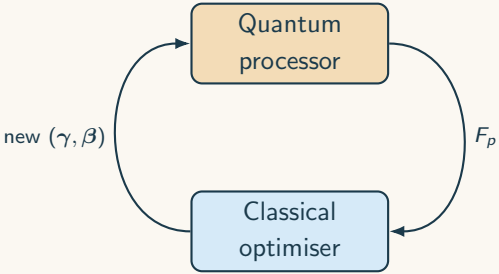
Implementation

The hybrid loop

Each iteration:

1. Prepare $|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$ on the QPU.
2. Measure $F_p = \langle H_C \rangle$ via repeated sampling.
3. Send F_p to a classical optimiser (COBYLA, L-BFGS-B, SPSA ...).
4. Receive new $(\boldsymbol{\gamma}, \boldsymbol{\beta})$, repeat until converged.

Only *cost values* (and optionally parameter-shift gradients) flow back — the quantum state never leaves the QPU.



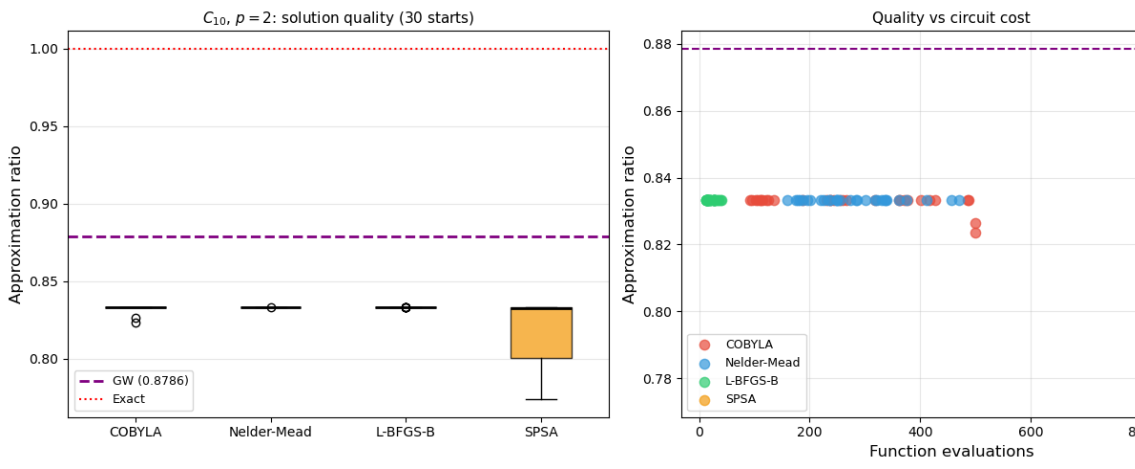
Choosing the classical optimiser

Optimiser	Gradient	Cost / step	Best for
COBYLA	none (trust-region)	$O(p)$	shot-noise robust
L-BFGS-B	parameter-shift	$2(\lvert E\rvert + n)p$	exact gradient (sim)
Nelder–Mead	none (simplex)	$O(p)$	fallback
SPSA	stochastic 2-eval	2	real hardware

Our choice: L-BFGS-B (primary) + COBYLA (cross-check)

L-BFGS-B converges fastest on noiseless statevector via the exact per-edge parameter-shift gradient. Running COBYLA in parallel as a gradient-free baseline rules out optimiser-specific artefacts. SPSA would replace both on real hardware.

Optimiser comparison

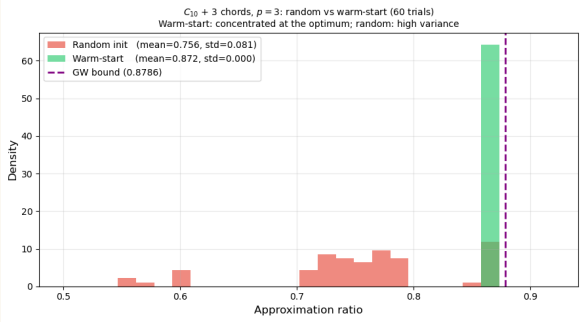


Warm start: initialise $p{+}1$ from the p-layer optimum

Algorithm (Zhou et al. 2020):

1. Optimise $p{=}1$ globally (random restarts).
2. For $k = 2, \ldots, p$:
   $\theta_0 \leftarrow (\boldsymbol{\gamma}^\star, \gamma^\star_{p-1}) \,\|\, (\boldsymbol{\beta}^\star, \beta^\star_{p-1}) + \eta$
   Local optimise F_k from θ_0.
3. Return final $(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)$.

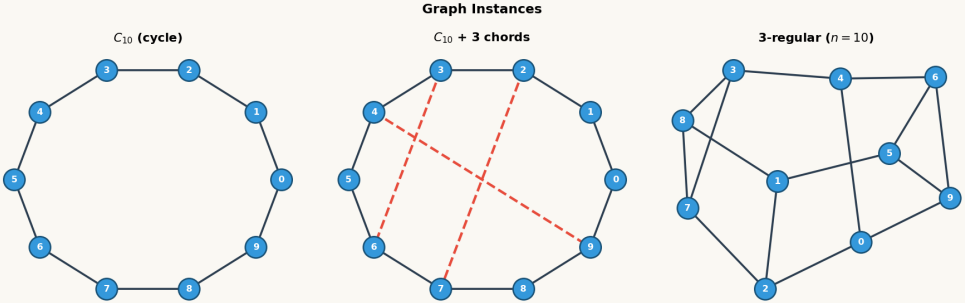
Intuition. A p-layer optimum is already a good region of parameter space for $p{+}1$ layers — the extra layer with small γ, β is nearly the identity.



Warm-start collapses the variance: std drops from 0.083/0.126 to 0.000.

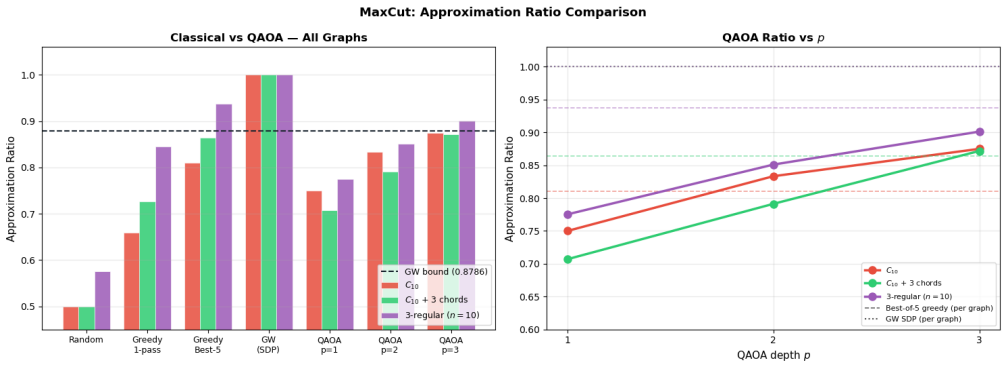
Results

Problem instances

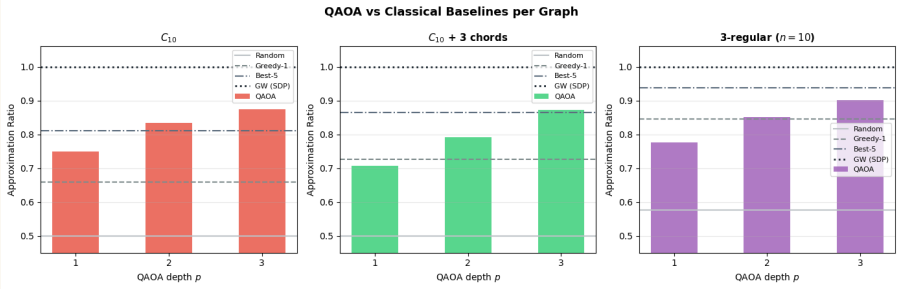


Graph	$\lvert E \rvert$	MaxCut	CX / layer
C_{10} (cycle)	10	10	20
C_{10} + 3 chords	13	13	26
3-regular ($n = 10$)	15	13	30

Performance comparison

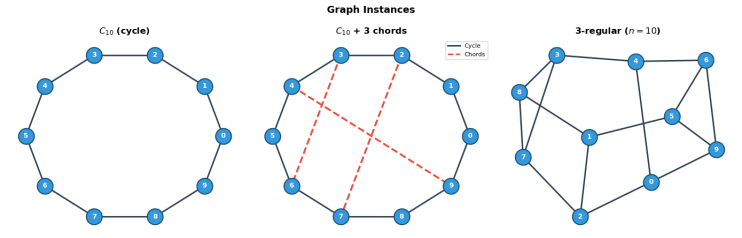


Approximation ratio versus depth



For all three graphs, $p = 3$ approaches the GW bound (0.8786).

Measurement distributions



C_{10}:
$|1010101010\rangle$
$|0101010101\rangle$

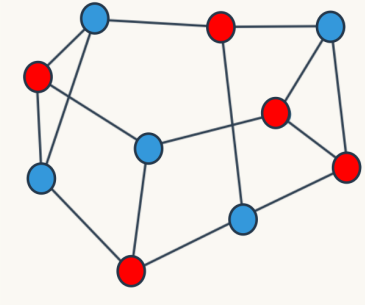
C_{10} + **3 chords:**
same two states

3-regular:
$|1100110100\rangle$
$|0011001011\rangle$

Two solutions per graph because every cut has a $\mathbb{Z}_2$ complement-symmetry $z \mapsto \bar{z}$.

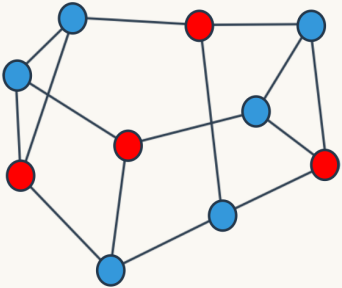
3-regular: not just one peak

MaxCut $= 13$



$|1100110100\rangle$, $|0011001011\rangle$

2nd best: cut $= 12$



$|0101101101\rangle$, $|1010010010\rangle$

Cut-12 partitions appear as the second-tallest peaks in the 3-regular histogram — QAOA concentrates amplitude on *good* cuts, not only the optimum.

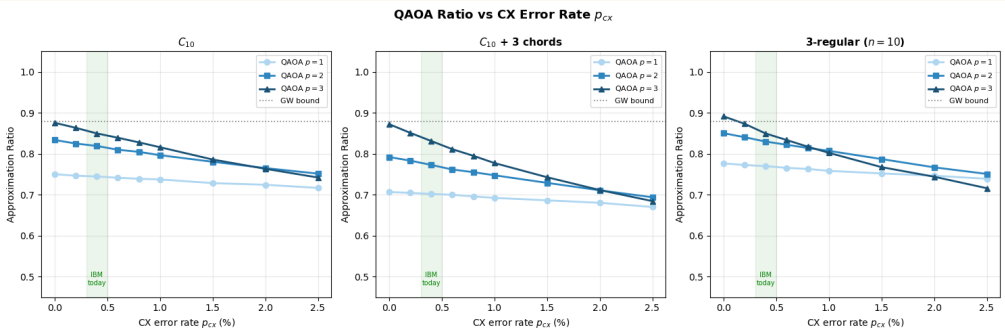
Introducing noise

- After every CX, apply a 2-qubit depolarising error with probability p_{cx}.
- Single-qubit gates (H, R_X, R_Z) get $p_{cx}/10$: 1Q fidelities on real hardware are typically $10\times$ better than 2Q.
- Total CX count $= 2p|E|$, so circuit fidelity decays as $(1 - p_{cx})^{2p|E|}$ — noise grows with depth and graph size.

Sweep range

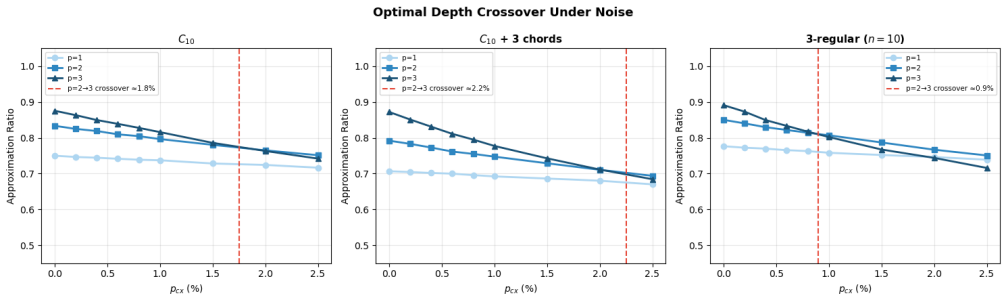
$p_{cx} \in [0, 0.025]$. IBM/Google today: $p_{cx} \approx 0.5$–1%. Fault-tolerant target: $p_{cx} \to 0$.

Ratio degradation with CX error rate



Each curve uses the noiseless-optimal (γ, β) from depth p.

Crossover: when shallower beats deeper



- Total CX count $= 2p|E|$, so circuit fidelity decays as $(1 - p_{cx})^{2p|E|}$.
- Above $p_{cx} \approx 1\%$, $p{=}1$ **outperforms** $p{=}3$ on every graph tested.
- Reported median CX errors on superconducting hardware sit in the 0.5–1% range — close to the crossover region.

References i

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Thanks for your attention!

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